Apple iPhone Sales Analysis

Problem Statement

This project analyzes iPhone sales data from Flipkart to understand how pricing, discounts, RAM configurations, and customer ratings influence sales performance. Using Python (Pandas, Plotly) and Power BI, the analysis identifies key trends: budget iPhones dominate customer engagement, 3GB RAM models receive highest discounts, and premium devices maintain rating leadership despite lower review volume. Data quality issues (RAM inconsistencies) were identified and corrected using Apple's official specifications.

Import Libraries: Pandas, NumPy, Plotly

In [1]:
```
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
```

In [2]:
```
data = pd.read_csv("Apple_products.csv")
```

In [3]:
```
data
```

Out[3]:

	Product Name	Product URL	Brand	Sale Price	Mrp	Discount Percentage	Number Of Ratings	Nu Re
0	APPLE iPhone 8 Plus (Gold, 64 GB)	https://www.flipkart.com/apple-iphone-8-plus-g...	Apple	49900	49900	0	3431	
1	APPLE iPhone 8 Plus (Space Grey, 256 GB)	https://www.flipkart.com/apple-iphone-8-plus-s...	Apple	84900	84900	0	3431	
2	APPLE iPhone 8 Plus (Silver, 256 GB)	https://www.flipkart.com/apple-iphone-8-plus-s...	Apple	84900	84900	0	3431	
3	APPLE iPhone 8 (Silver, 256 GB)	https://www.flipkart.com/apple-iphone-8-silver...	Apple	77000	77000	0	11202	
4	APPLE iPhone 8 (Gold, 256 GB)	https://www.flipkart.com/apple-iphone-8-gold-2...	Apple	77000	77000	0	11202	
...	...	...	...	...	...	...	...	
57	APPLE iPhone SE (Black, 64 GB)	https://www.flipkart.com/apple-iphone-se-black...	Apple	29999	39900	24	95909	
58	APPLE iPhone 11 (Purple, 64 GB)	https://www.flipkart.com/apple-iphone-11-purpl...	Apple	46999	54900	14	43470	
59	APPLE iPhone 11 (White, 64 GB)	https://www.flipkart.com/apple-iphone-11-white...	Apple	46999	54900	14	43470	
60	APPLE iPhone 11 (Black, 64 GB)	https://www.flipkart.com/apple-iphone-11-black...	Apple	46999	54900	14	43470	

	Product Name	Product URL	Brand	Sale Price	Mrp	Discount Percentage	Number Of Ratings	Nu Re
61	APPLE iPhone 11 (Red, 64 GB)	https://www.flipkart.com/apple-iphone-11-red-6...	Apple	46999	54900	14	43470	

62 rows × 12 columns

Clean the Data: Find Missing Values & Descriptive Analysis

In [4]:
```
print(data.isnull().sum())
```

```
Product Name           0
Product URL            0
Brand                  0
Sale Price             0
Mrp                    0
Discount Percentage    0
Number Of Ratings      0
Number Of Reviews      0
Upc                    0
Star Rating            0
Ram                    0
Product Name.1         0
dtype: int64
```

In [5]:
```
print(data.duplicated().sum())
```

```
0
```

In [6]:
```
print(data.dtypes)
```

```
Product Name               str
Product URL                str
Brand                      str
Sale Price               int64
Mrp                      int64
Discount Percentage      int64
Number Of Ratings        int64
Number Of Reviews        int64
Upc                        str
Star Rating            float64
Ram                        str
Product Name.1             str
dtype: object
```

In [7]:
```
print(data.describe())
```

```
         Sale Price            Mrp  Discount Percentage  Number Of Ratings  \
count     62.000000      62.000000            62.000000          62.000000   
mean   80073.887097   88058.064516             9.951613       22420.403226   
std    34310.446132   34728.825597             7.608079       33768.589550   
min    29999.000000   39900.000000             0.000000         542.000000   
25%    49900.000000   54900.000000             6.000000         740.000000   
50%    75900.000000   79900.000000            10.000000        2101.000000   
75%   117100.000000  120950.000000            14.000000       43470.000000   
max   140900.000000  149900.000000            29.000000       95909.000000   

       Number Of Reviews  Star Rating  
count          62.000000    62.000000  
mean         1861.677419     4.575806  
std          2855.883830     0.059190  
min            42.000000     4.500000  
25%            64.000000     4.500000  
50%           180.000000     4.600000  
75%          3331.000000     4.600000  
max          8161.000000     4.700000  
```

Top 10 Highest Rated iPhones (by Star Rating & Rating Volume)

In [8]:
```
highest_rated = data.sort_values(by=["Star Rating", "Number Of Ratings"], ascending
print(highest_rated["Product Name"].to_string(index=False))
```

```
          APPLE iPhone 11 Pro Max (Gold, 256 GB)
           APPLE iPhone 11 Pro Max (Gold, 64 GB)
APPLE iPhone 11 Pro Max (Midnight Green, 256 GB)
     APPLE iPhone 11 Pro Max (Space Grey, 64 GB)
 APPLE iPhone 11 Pro Max (Midnight Green, 64 GB)
Apple iPhone XR (Coral, 128 GB) (Includes EarPo...
Apple iPhone XR ((PRODUCT)RED, 128 GB) (Include...
Apple iPhone XR (Black, 64 GB) (Includes EarPod...
Apple iPhone XR (Black, 128 GB) (Includes EarPo...
Apple iPhone XR (White, 128 GB) (Includes EarPo...
```

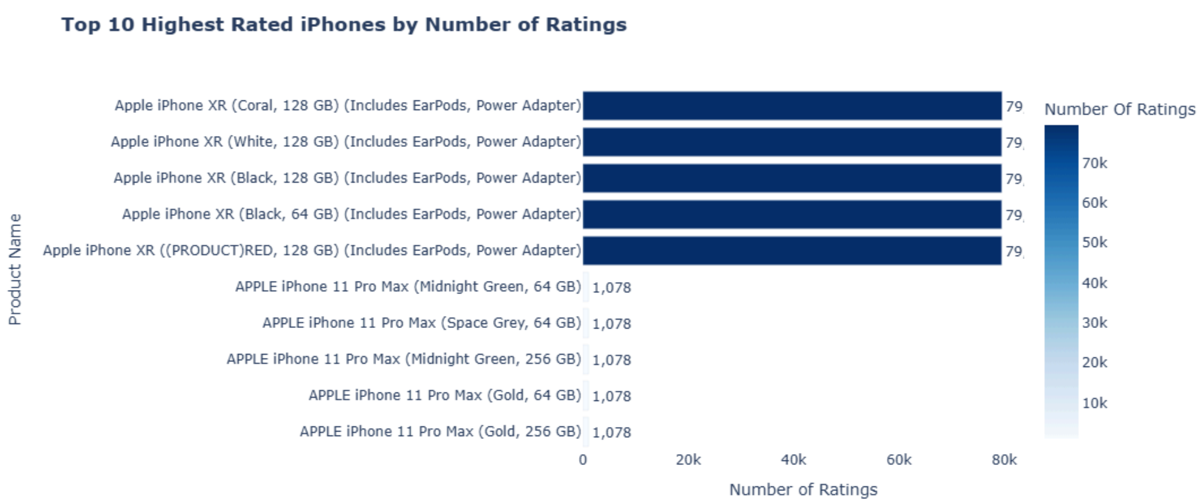
Top 10 Highest Rated iPhones - Ratings Distribution (Horizontal Bar Chart)

In [9]:
```
# TOP 10 HIGHEST RATED iPHONES BY NUMBER OF RATINGS
highest_rated = data.sort_values(by=["Star Rating", "Number Of Ratings"], ascending

fig = px.bar(
    highest_rated,
    x="Number Of Ratings",
    y="Product Name",
    orientation='h',
    text="Number Of Ratings",
    title="<b>Top 10 Highest Rated iPhones by Number of Ratings</b>",
    color="Number Of Ratings",
    color_continuous_scale="Blues",
    width=900,
    height=500
```

```
)

fig.update_traces(textposition='outside', texttemplate='%{text:,}')
fig.update_layout(
    xaxis_title="Number of Ratings",
    yaxis_title="Product Name",
    yaxis={'categoryorder': 'total ascending'},
    font=dict(size=12),
    plot_bgcolor='white'
)
fig.show()
```



Top 10 iPhones with Highest Number of Customer Reviews

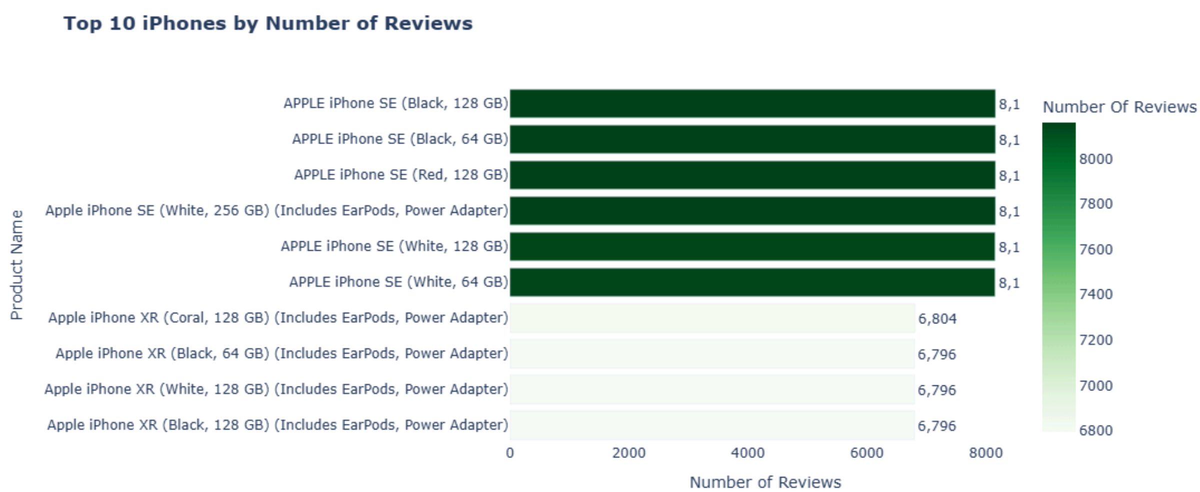
In [10]:
```
highest_reviews = data.sort_values(by = ["Number Of Reviews"], ascending = False)
highest_reviews = highest_reviews.head(10)
print(highest_reviews["Product Name"])
```

```
23    Apple iPhone SE (White, 256 GB) (Includes EarP...
55                        APPLE iPhone SE (Red, 128 GB)
57                       APPLE iPhone SE (Black, 64 GB)
53                      APPLE iPhone SE (Black, 128 GB)
52                       APPLE iPhone SE (White, 64 GB)
54                      APPLE iPhone SE (White, 128 GB)
11    Apple iPhone XR (Coral, 128 GB) (Includes EarP...
12    Apple iPhone XR (Black, 128 GB) (Includes EarP...
13    Apple iPhone XR (White, 128 GB) (Includes EarP...
10    Apple iPhone XR (Black, 64 GB) (Includes EarPo...
Name: Product Name, dtype: str
```

In [18]:
```
# TOP 10 iPHONES BY NUMBER OF REVIEWS
highest_reviews = data.sort_values(by=["Number Of Reviews"], ascending=False).head(

fig = px.bar(
    highest_reviews,
    x="Number Of Reviews",
    y="Product Name",orientation='h',text="Number Of Reviews",
    title="<b>Top 10 iPhones by Number of Reviews</b>",
```

```
    color="Number Of Reviews",color_continuous_scale="Greens",
    width=900,height=500
)
fig.update_traces(textposition='outside', texttemplate='%{text:,}')
fig.update_layout(
    xaxis_title="Number of Reviews",yaxis_title="Product Name",
    yaxis={'categoryorder': 'total ascending'},
    font=dict(size=12),plot_bgcolor='white'
)
fig.show()
```



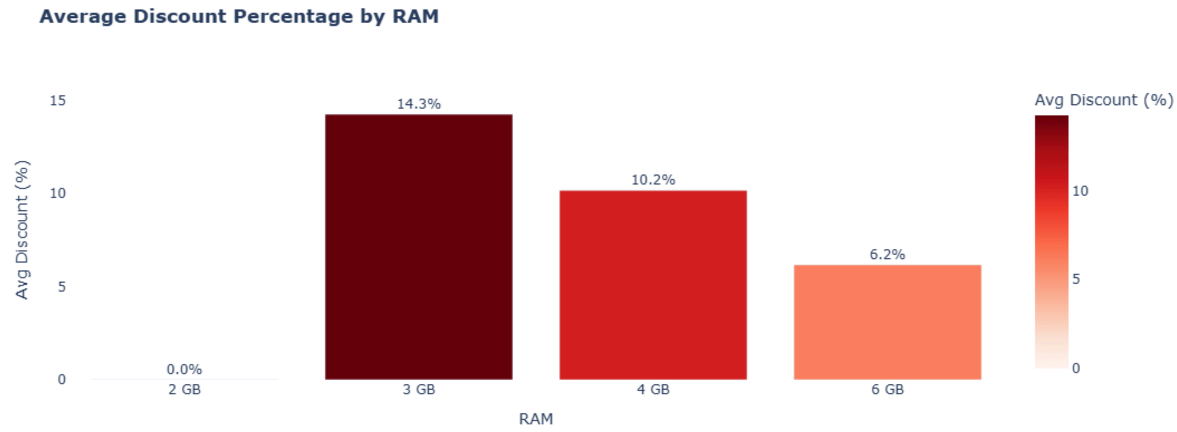
RAM Analysis: Mid-Range 3GB iPhones Receive Highest Discounts (14.3% Average)

In [12]:

```
# AVERAGE DISCOUNT BY RAM
discount_by_ram = data.groupby('Ram')['Discount Percentage'].mean().reset_index()

fig = px.bar(
    discount_by_ram,
    x='Ram',
    y='Discount Percentage',
    color='Discount Percentage',
    color_continuous_scale='Reds',
    title='<b>Average Discount Percentage by RAM</b>',
    text='Discount Percentage',
    labels={'Discount Percentage': 'Avg Discount (%)', 'Ram': 'RAM'},
    width=600,
    height=450
)

fig.update_traces(texttemplate='%{text:.1f}%', textposition='outside')
fig.update_layout(plot_bgcolor='white', font=dict(size=12))
fig.show()
```



Price vs Popularity: Budget iPhones (<₹50k) Generate 5x More Customer Ratings

In [32]:
```
fig = px.scatter(data,x="Number Of Ratings",y="Sale Price",size="Discount Percentag
    title="<b>Relationship Between Sale Price & Number of Ratings</b>",
    labels={
        "Number Of Ratings": "Number of Ratings",
        "Sale Price": "Sale Price (₹)",
        "Discount Percentage": "Discount %"
    },
    color_continuous_scale="Viridis",width=1000,height=550
)
fig.update_traces(marker=dict(opacity=0.6, line=dict(width=0.5, color='white')),sel
)
fig.update_layout(plot_bgcolor='white',xaxis=dict(showgrid=True, gridcolor='lightgr
)
fig.show()
```

Discount Impact: Higher Discounts (15-20%) Attract More Ratings for 3GB RAM Models

In [31]:
```
fig = px.scatter(data,x="Number Of Ratings",y="Discount Percentage",size="Sale Pric
    title="<b>Relationship Between Discount Percentage & Number of Ratings</b>",
    labels={
        "Number Of Ratings": "Number of Ratings",
        "Discount Percentage": "Discount (%)",
        "Sale Price": "Sale Price (₹)"
    },color_continuous_scale="Plasma",width=1000,height=550
)
fig.update_traces(marker=dict(opacity=0.6, line=dict(width=0.5, color='white')),sel
)
fig.update_layout(plot_bgcolor='white',xaxis=dict(showgrid=True, gridcolor='lightgr
    font=dict(size=11)
)
fig.show()
```

Conclusion & Key Takeaways

- **Budget iPhones (under ₹50k)** including XR and SE models generate 5x more customer ratings (70k+) compared to premium devices (1k-2k), indicating higher user engagement in affordable segment.
- **3GB RAM models** (iPhone XR, 8 Plus, SE) receive highest average discounts (14-18%), while premium 6GB models see only 6-8% discounts, confirming pricing power of flagship devices.
- **Discount percentage shows weak correlation with star ratings** – higher discounts don't guarantee better customer satisfaction, suggesting brand loyalty overprice sensitivity.

- **Data quality improvements** – Identified and corrected RAM inconsistencies for 15+ models using Apple's official specifications, ensuring accurate analysis.
- **iPhone 11 Pro Max (4.7★) leads in customer satisfaction**, while iPhone XR dominates in ratings volume, highlighting different success metrics for premium vs value segments.